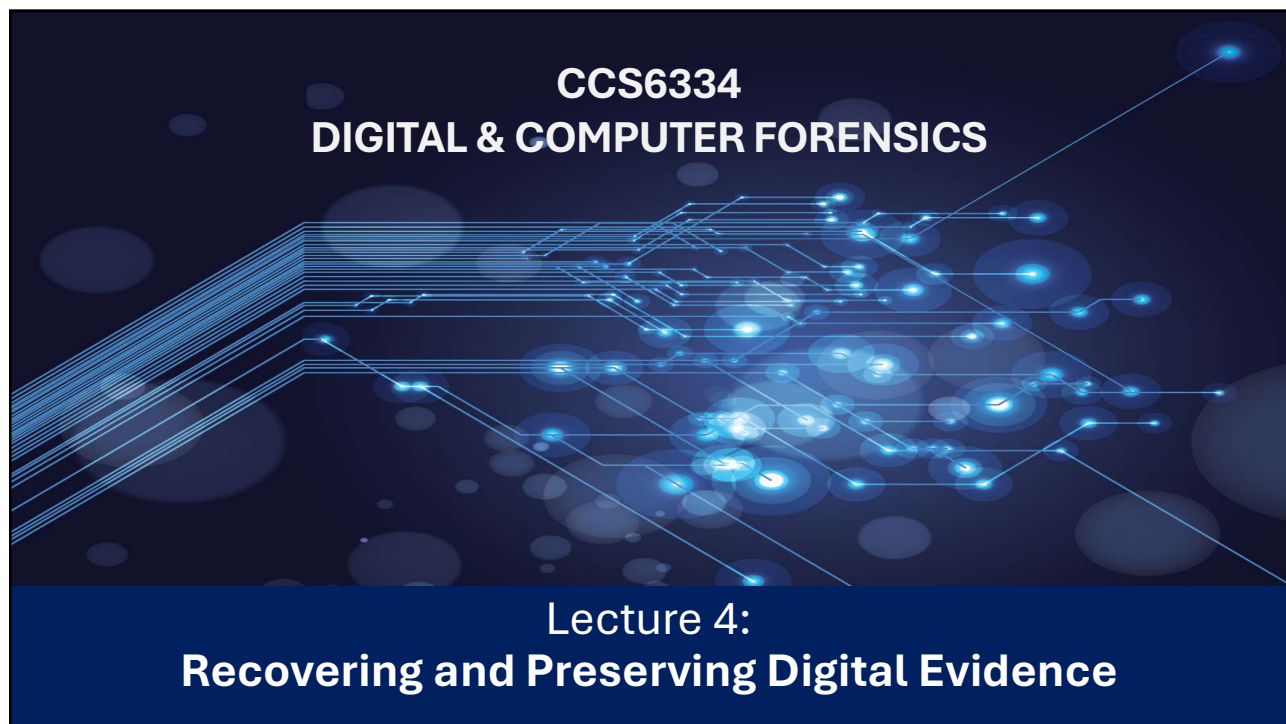




1



## Learning Objectives

After studying this chapter, you should be able to:

- Understanding the chain of custody
- Describing the physical acquisition and safekeeping
- Recovering digital evidence through forensic imaging
- Acquiring digital evidence through live recovery



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# Chain of Custody



- **Purpose:** Protect the integrity, authenticity, and reliability of evidence throughout the investigation and legal process.
- **Key Requirement:** Evidence must be collected, handled, and stored with extreme care to prevent **contamination, tampering, or alteration**—whether intentional or accidental.
- **Legal Impact:**
  - Contaminated or altered evidence may be ruled **inadmissible**.
  - Opposing counsel can challenge tainted evidence, casting **doubt on authenticity**.
  - In criminal cases, doubt may favour the defendant, reducing the value of the evidence.
- **Court Expectations:**
  - Strict adherence to **collection procedures** and **chain-of-custody documentation**.
  - Evidence must remain protected and unchanged from **acquisition to presentation**.
  - If any alteration occurred, it must be **fully disclosed and justified** to the court.



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# Why Safeguard Evidence?



**The Core Objective:** Prevent intentional or unintentional tampering to maintain evidence in a pristine condition.



## Integrity is Paramount

Any challenge from opposing counsel regarding contamination can lead to evidence being struck out.



## Risk of Doubt

If evidence is tainted, its value diminishes.

Juries struggle to determine reliability.

Judges may direct benefit of the doubt to the defendant.



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## Acquisition & Safekeeping of Digital Evidence



- High Vulnerability – Digital evidence can be easily contaminated or damaged during collection/extraction
- Safe Custody – Proper record-keeping and secure storage required for physical and digital records
- Court Expectations – Evidence must remain unaltered from seizure to presentation in legal proceedings
- Crime Scene Risks – Weather, animals, and human interference can compromise evidence integrity
- Digital Crime Scene – Must be protected from contamination and unauthorized access
- Untrained Personnel – Well-intentioned actions (e.g., supervisors previewing data) risk altering or invalidating evidence
- Need for Training – Proper forensic skills essential to preserve authenticity and reconstruct events

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## Chain of Custody in Digital Evidence



- ⚠️ Complex Process – Requires great care to avoid contamination, destruction, or loss
- 🚫 Common Pitfalls – Inexperienced examiners may unintentionally alter metadata
- 📦 Secure Storage Alone Not Enough – Formal records of exhibit history are essential
- ⚖️ Court Expectations – Same assurances as physical evidence; documented careful handling required
- 📄 Detailed Records – Each exhibit and forensic image must include:
  - Full description of exhibit
  - Case reference and custodian details
  - Signatures of all parties involved
- ✅ Outcome – Enhances confidence in authenticity and genuineness of digital evidence

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[illegible]

| EVIDENCE CHAIN OF CUSTODY TRACKING FORM For Digital Evidence |  |  |
|--------------------------------------------------------------|--|--|
| Matter Number : _____ Client Number : _____                  |  |  |
| Original Submitting Person (Name/position) _____             |  |  |
| Employee or Representative For Respondent: _____             |  |  |
| Employee or Representative For Plaintiff: _____              |  |  |
| Date/Time Acquired : _____                                   |  |  |
| Address Acquired _____                                       |  |  |
| GPS location Acquired _____                                  |  |  |

| Description of Evidence |          |                                                                                            |
|-------------------------|----------|--------------------------------------------------------------------------------------------|
| Item #                  | Quantity | Description of Item (Model, Serial #, Condition)<br>Photographs as Evidence attached _____ |
|                         |          |                                                                                            |
|                         |          |                                                                                            |
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| Chain of Custody |           |                                  |                                  |                   |
|------------------|-----------|----------------------------------|----------------------------------|-------------------|
| Item #           | Date/Time | Released by<br>(Signature & ID#) | Received by<br>(Signature & ID#) | Comments/Location |
|                  |           |                                  |                                  |                   |
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| Chain of Custody |           |                                  |                                  |                   |
|------------------|-----------|----------------------------------|----------------------------------|-------------------|
| Item #           | Date/Time | Released by<br>(Signature & ID#) | Received by<br>(Signature & ID#) | Comments/Location |
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| Final Disposal Authority                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Authorization for Disposal</b><br>Item(s) #: _____ on this document pertaining to matter(s) _____<br>is/are no longer needed as evidence and is/are authorized for disposal by [check appropriate disposal method]<br><input type="checkbox"/> Return to Owner <input type="checkbox"/> Destroy/Divert<br>Name of Authorizing person: _____ Signature: _____ Date: _____                                                                                                              |  |
| <b>Witness to Destruction of Evidence</b><br>Item(s) #: _____ on this document were destroyed by Evidence Custodian _____ ID#: _____<br>in my presence on (Date) _____<br>Name & ID# of Witness to destruction: _____ Signature: _____<br>Date: _____                                                                                                                                                                                                                                    |  |
| <b>Release to Lawful Owner</b><br>Item #: _____ on this document was/were released by Evidence Custodian _____ ID#: _____ to _____<br>Name _____<br>Address: _____ City: _____ State: _____ Zip Code: _____<br>Telephone Number: (____) _____<br>Under penalty of law, I certify that I am the lawful owner of the above item(s).<br>Signature: _____ Date: _____<br>Copy of Government-issued photo identification is attached <input type="checkbox"/> Yes <input type="checkbox"/> No |  |
| This Evidence Chain-of-Custody form is to be retained as a permanent record                                                                                                                                                                                                                                                                                                                                                                                                              |  |

A sample chain of custody form

## Seizure and Initial Inspection of Digital Devices



- **Greater Impact of Inadmissibility:** Digital evidence is highly interconnected; losing a single artifact can significantly weaken a case.
- **Need for Corroboration:** Practitioners should link or correlate digital artifacts to support timelines and hypotheses.
- **Risk of Tampering:** Any alteration—before, during, or after recovery—can modify data/metadata (e.g., timestamps) and reduce evidential value.
- **Preserving Integrity:** Secure and careful handling of seized devices is essential to maintain authenticity.
- **Forensic Imaging Procedure:**
  - Carefully remove the hard drive; avoid physical damage.
  - Use sterile antistatic gloves to prevent short circuits.
  - Tag and place both the original drive and imaging drive in antistatic bags with full details (date/time, case ID, device info, serial numbers, practitioner name).
- **Documentation & Chain of Custody:**
  - Duplicate and cross-reference exhibit information in the custody register.
  - Record acquisition details, location, and all forensic tools/methods used.

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## Chain of Custody – Handling & Preservation



- Custodian must demonstrate unbroken chain of custody from record creation to court
- Handle original devices minimally; document and justify any alterations
- Recovery/preservation aims to stabilize the digital crime scene and minimize data loss
- Standard procedures:
  - Photograph/video crime scene and monitor displays
  - Record exhibit position, cabling, and powered state
  - Use antistatic gloves/bands to prevent hardware damage during drive removal

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## Chain of Custody – Handling & Preservation



- Disconnect interfering telecom/Bluetooth devices; consider **Faraday cages** for mobile phones
- Transport evidence safely: protect from heat, moisture, dust, shocks; tag clearly to avoid misidentification
- Store evidence securely until examination; physical devices often link digital evidence to suspects
- Apply **Locard's principle**: collect DNA/fingerprints to establish access to devices
- Practitioner must verify integrity in court, often by **hash comparison** of forensic image vs. original
- Digital evidence preserved on secure storage media; strict forensic processes required before copying

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## RECOVERING DIGITAL EVIDENCE THROUGH FORENSIC IMAGING



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## Possession & Recovery of Digital Evidence



- Physical exhibits with digital evidence must be stored securely, like other case exhibits
- Courts recognize high risk of tampering and authentication problems with digital evidence
- Reliable forensic tools help minimize contamination during recovery
- Digital forensics is still evolving, with ongoing debate over standards and processes
- Practitioners may choose dead recovery (preferred) or live recovery methods
- Dead recovery involves creating a forensic image without activating the operating system
- This avoids contamination and copying errors, ensuring admissibility in court
- Poor practices during imaging can compromise evidence integrity and admissibility



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## Dead Analysis Evidence Recovery



- **Definition:** Forensic copying of data without using the computer's operating system
- **"Dead" State:** OS remains inactive; recovery uses trusted CD/USB or extracted hard drive
- **Methods:**
  - Power down device → remove drive → image separately
  - Power on via external boot media → copy drive in situ
- **Tools:** Hardware write blockers and trusted forensic applications ensure integrity
- **Purpose:** Prevents OS processes from altering data, avoids contamination and copying errors
- **Outcome:** Reliable forensic image created for analysis, maintaining evidentiary value

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## Hardware Write Blockers

- Dead recovery terminates OS processes to prevent alteration of evidence
- Write-blocker hardware/software prevents overwriting during forensic imaging
- Ensures forensic images remain unaltered for future analysis in trusted environments
- Forensic images can be duplicated for backup, examiner review, or independent inspection
- Best practice: collect only necessary data, avoid indiscriminate imaging of entire drives
- Risk: normal computer operations may unintentionally write to evidence devices
- Solution: use write-blockers to allow safe data recovery without contamination
- Examples of approved HW write blockers :

<https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt/cftt-technical/hardware>



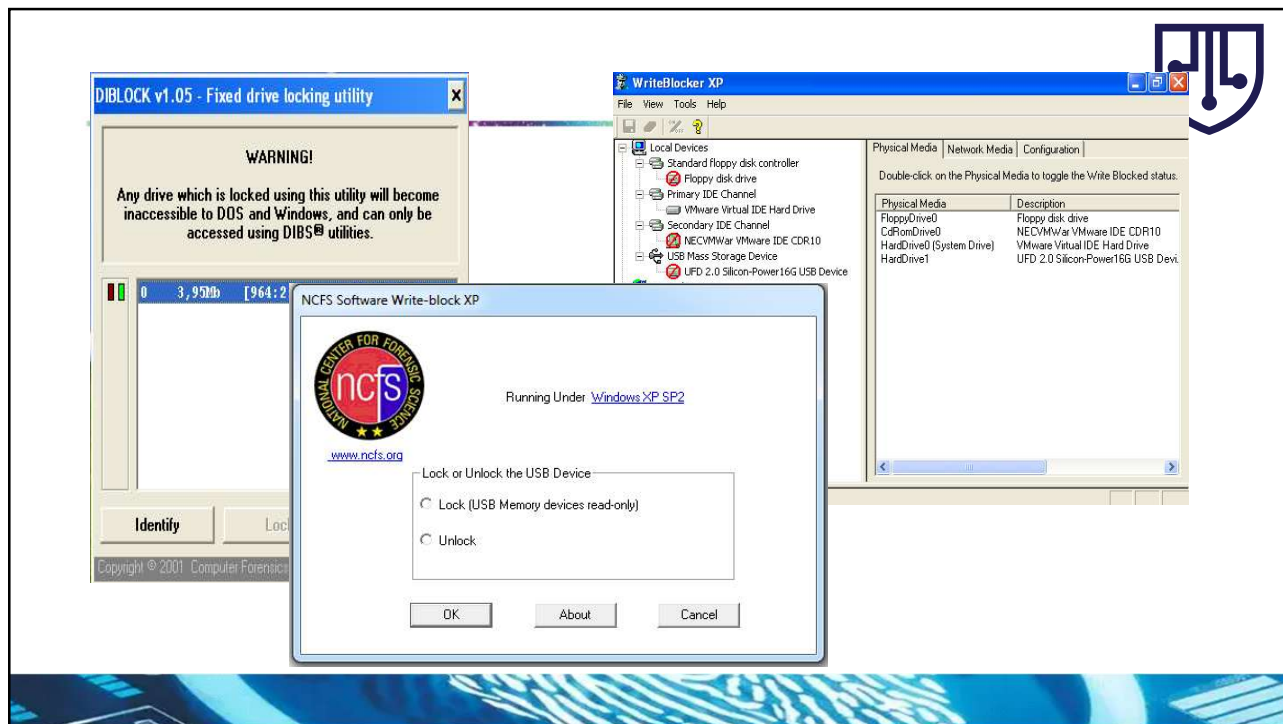
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## Software Write Blockers

- Similar to hardware write blockers but run as an application instead – the application is run on the operating system that implements a software control to turn off the write capability of the operating system
- Advantage – no additional/specialized hardware, more upgradeable and flexible, might not require physical connection to source
- Disadvantage – difficult to explain and justify operations performed, reliant on underlying hardware/software application is running on
- Examples of approved SW write blockers:
  - <https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt/cftt-technical/software>

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## Enhancing Data Preservation During Recovery



- **Secure Possession:** Physical exhibits containing digital evidence are stored like other evidence to prevent tampering.
- **Legal Recognition:** Courts acknowledge high risk of tampering and authentication issues with digital evidence.
- **Forensic Tools:** Reliable tools minimize contamination during data recovery; standards are still evolving in digital forensics.
- **Recovery Approaches:**
  - **Dead Recovery:** Forensic imaging without powering the device or activating the OS.
  - **Live Recovery:** Accessing data while the system is powered on (higher risk).
- **Forensic Imaging:**
  - Preferred method for **dead analysis**.
  - Avoids contamination, preserves data integrity, and ensures admissibility.
- **Risks:** Poor practices during imaging can jeopardize evidence admissibility.

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## Drawbacks of Dead Recovery



- Dead recovery may block access to volatile memory, but remnants can remain in file slack
- RAM slack: unused cluster space filled with random memory data, recoverable for recent/old events
- Drive slack: extra sector space padded with remnants of deleted data
- Slack areas may contain valuable information such as logon names, passwords, and activity traces
- Recovery often requires advanced data carving during forensic image analysis
- Live recovery provides easier access to volatile memory, but with higher contamination risks

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## RECOVERING DIGITAL EVIDENCE THROUGH LIVE RECOVERY



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## Live Recovery of Digital Evidence



- Involves capturing volatile memory, system processes, and network data before powering down
- Conducted while the operating system is still running
- Traditionally, dead recovery was preferred to avoid accidental deletion or contamination
- Courts accepted forensic images of hard drives as reliable evidence
- Risks of live recovery:
  - Data may be overwritten or erased during examination
  - OS/software may be modified to delete or falsify information
- Used only in exceptional circumstances when original data must be accessed
- Practitioners must be competent, justify their actions, and provide a full recovery report

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## Benefits of Live Recovery



- Produces a snapshot of targeted system areas (not full bit-for-bit image)
- Snapshot may not be reproducible later if drive use modifies earlier data
- Possibility of obtaining image that bypasses encryption
- Running memory can also be captured, providing a complete picture of how the system was used immediately prior to the imaging process
- Can avoid any anti-forensics measures possibly applied
- Preserves and collects volatile data (RAM, processes, network info) before shutdown
- Maintains the state of the operating system and data at time of seizure
- Useful in incident-response triage to quickly assess significant events

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## Information from Live Recovery



- Some examples of information possibly recover-able in live recovery
  - System time/date
  - Network connections, open TCP/UDP ports and associated programs using the ports
  - Currently logged in users
  - Internal network routing table
  - Running processes and Scheduled jobs
  - Open files
- Info like this can help correlate information between the computer being examined and other network-linked computers
- Notice that unlike dead analysis, when performing live analysis, the data collected is often more targeted (e.g. tmp files only) and restricted to certain items (e.g. live memory dumps)

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## Live Recovery Tools



- Some examples (not exhaustive) of live recovery tools include
  - Memory dumping
    - Winen → now commercial only
    - WinHex → commercial
    - KnTTools → unsupported program, binaries available
    - Belkasoft Live RAM Capturer → free, closed source
  - Live imaging
    - FTK Imager → free, closed source
    - Encase Forensic Imager → free, closed source
    - Belkasoft Evidence Center X → commercial

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## Challenges of Live Recovery



- Many practitioners remain reluctant, viewing live recovery as unsound, though it may become more common
- Closing down devices risks loss of metadata, inaccessible encrypted drives, and loss of RAM data
- Authenticity and reliability challenged by anti-forensic toolkits
- Anti-forensic tactics may booby-trap devices, deleting or altering data when booted
- Newer enhanced forensic tools are being developed to address these problems, but raise further issues

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## Standards for Digital Forensic Tools



- Forensic tools assist in filtering large datasets, carving deleted files, and locating hidden data
- **ISO/IEC 27041:2015** provides guidelines for tool validation, but criticized as lacking depth and detail
- **NIST (USA)**: Computer Forensics Tool Testing ensures reliability, accuracy, and minimal contamination
- **UK Forensic Science Regulator (2014)**: Draft guidelines on method validation, requiring documentation of recovery processes
- **IOCE**: Advocates independent assessment or provider assurance; promotes cooperation among law enforcement for validation
- **Daubert Test (USA)**: Courts benchmark tool validity; require scientific assurances beyond manufacturer claims
- Challenges:
  - Lack of consistent standards and compliance requirements
  - Duplication of efforts, poor information sharing
  - Increasing difficulty of analyzing large datasets with limited automatio

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## Case Studies Involving Custody Chain



### CASE 1 (<https://bit.ly/3S0w30D>)

- Case against employee (school teacher) using company provided hardware was found to possess nude photos of school students
- Although company rules and policies state that they have rights to inspect and seize company owned equipment/data, personal privacy rights trumped the rules in court and important pieces of evidence was thrown

### CASE 2 (<https://jade.io/article/12808>)

- Criminal case of Colin Francis Mowday who was caught viewing and downloading child pornography in a public café in Merridin WA
- Due to lost in CoC, the evidence was thrown out of court

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## Summary



What was covered today:

- We looked at evidence recovery and preservation in a general sense and then focused on how these two interlinked requirements relate to digital evidence
- Dead and live evidence recovery processes have been described, although these well-established processes are now facing a paradigm shift heralded by enhanced technologies



Reference Source: Practical Digital Forensics (Packt Publishing)

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